

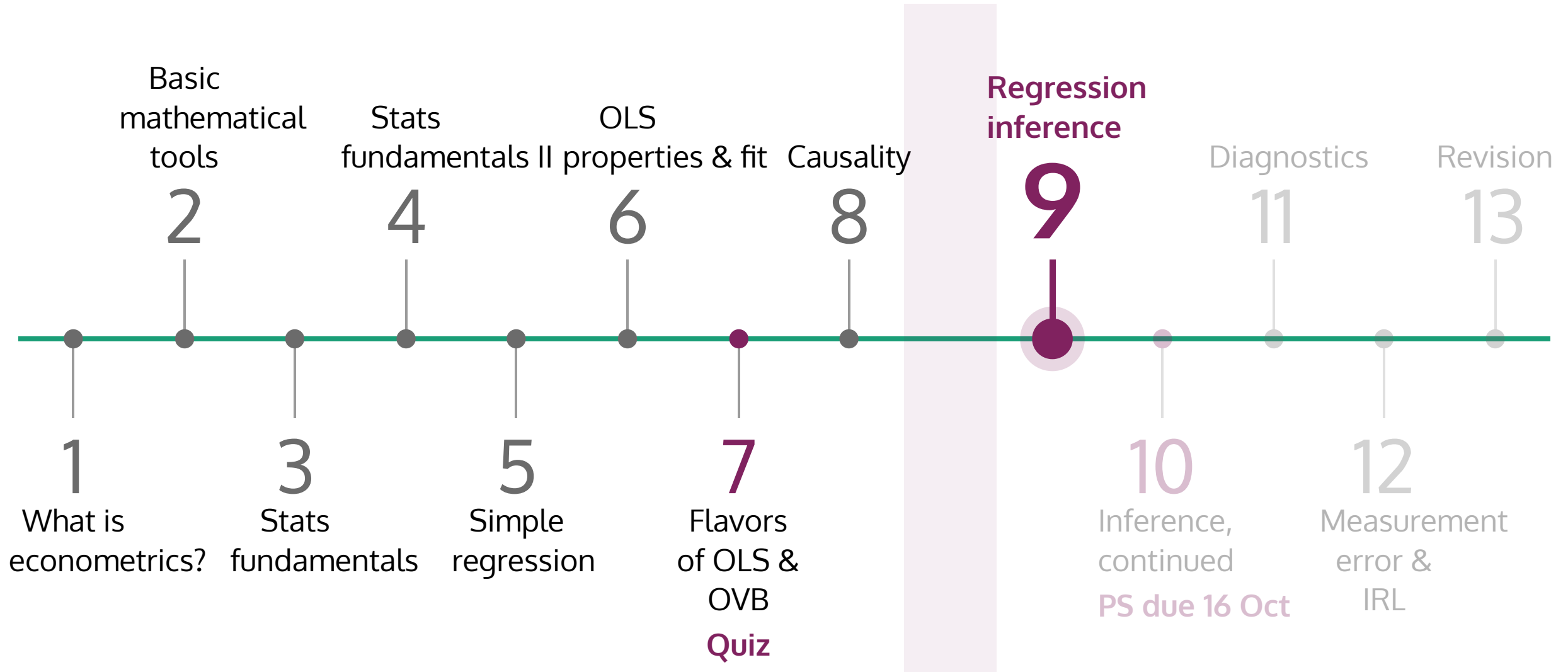


ECON 2041

Introductory Econometrics

How sure are we?

Plan for the semester



Plan for today

- The rest of the regression table
- Slope sampling distribution
- Testing null hypotheses
- Do the slopes differ?

LET & LEU

1. LET (my teaching): closes **Friday 9 October**, 11 p.m.
2. LEU (the unit): Monday 12 October to Sunday 1 November

Fewer datasets, used more times

Personalized feedback on the quiz

Practice questions like the quiz and the final exam

Related topics taught more closely together

The rest of the regression table

We judge how sure we are of the slope

```
model = ols("exp_per_adult ~ egm_per_1000", data=pokies).fit()
```

	coef	std err	t	P> t	[0.025	0.975]
Intercept	71.1722	38.992	1.825	0.072	-6.471	148.816
egm_per_1000	78.2968	7.205	10.866	0.000	63.949	92.645

You can follow along in a notebook

<https://emiliatjernstrom.com/econ2041/lectures/w09>

As usual, it will open directly in Colab

Click **Copy to Drive** first, so you can save your edits

Slope sampling distribution

In the population, we know the true slope

```
pop = ols("math ~ escs", data=pisa).fit()
```

	coef	std err	t	P> t	[0.025	0.975]
escs	43.8472	0.971	45.138	0.000	41.943	45.751

Slope estimate across samples

```
sample = pisa.sample(300, random_state=1)
ols("math ~ escs", data=sample).fit()
```

	coef	std err	t	P> t	[0.025	0.975]

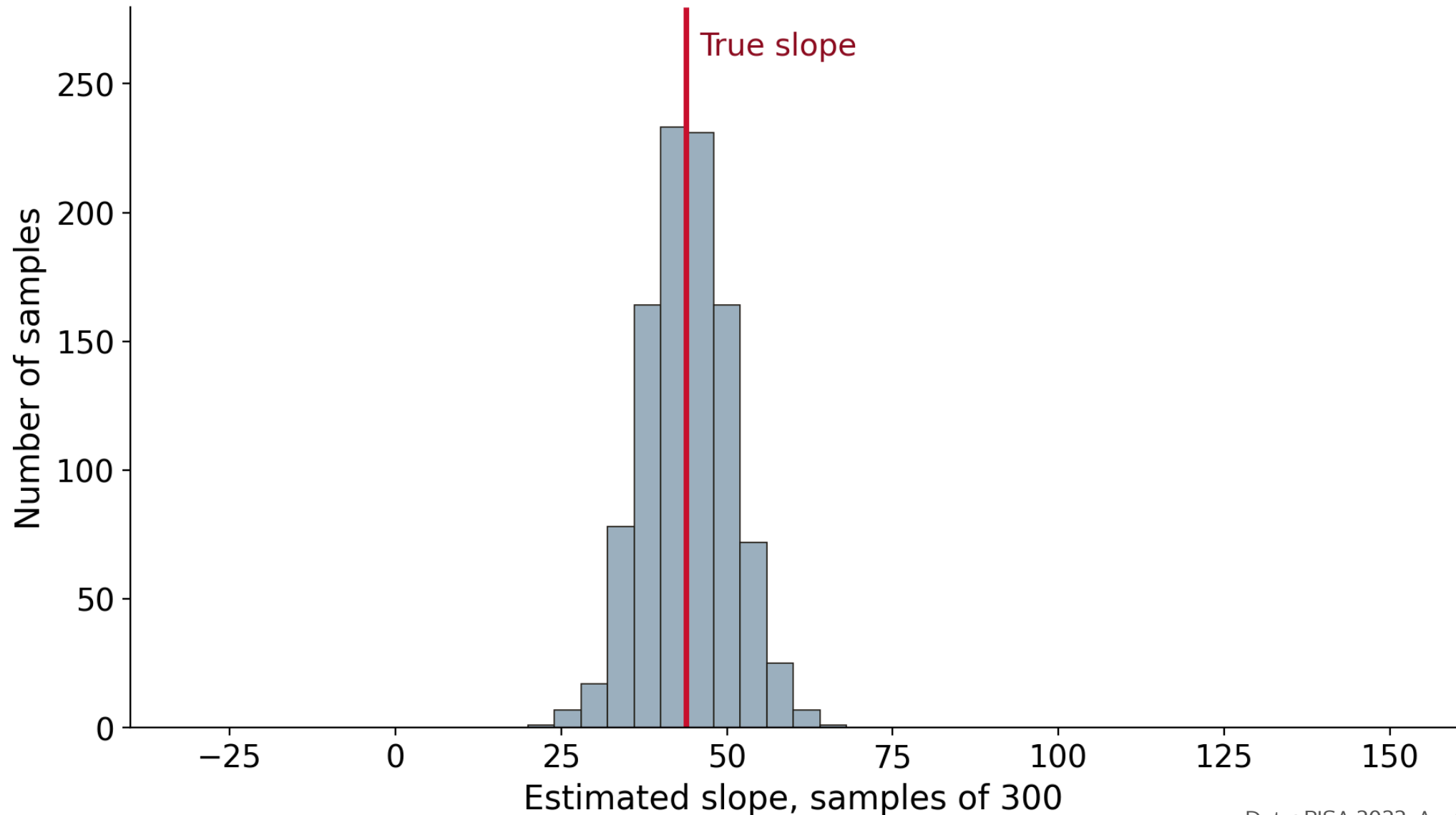
escs	42.0037	5.944	7.067	0.000	30.307	53.701

```
sample = pisa.sample(300, random_state=2)
ols("math ~ escs", data=sample).fit()
```

	coef	std err	t	P> t	[0.025	0.975]

escs	44.0509	6.166	7.144	0.000	31.916	56.186

Sample slopes are centered on the true slope



Data: PISA 2022, Australia (OECD)

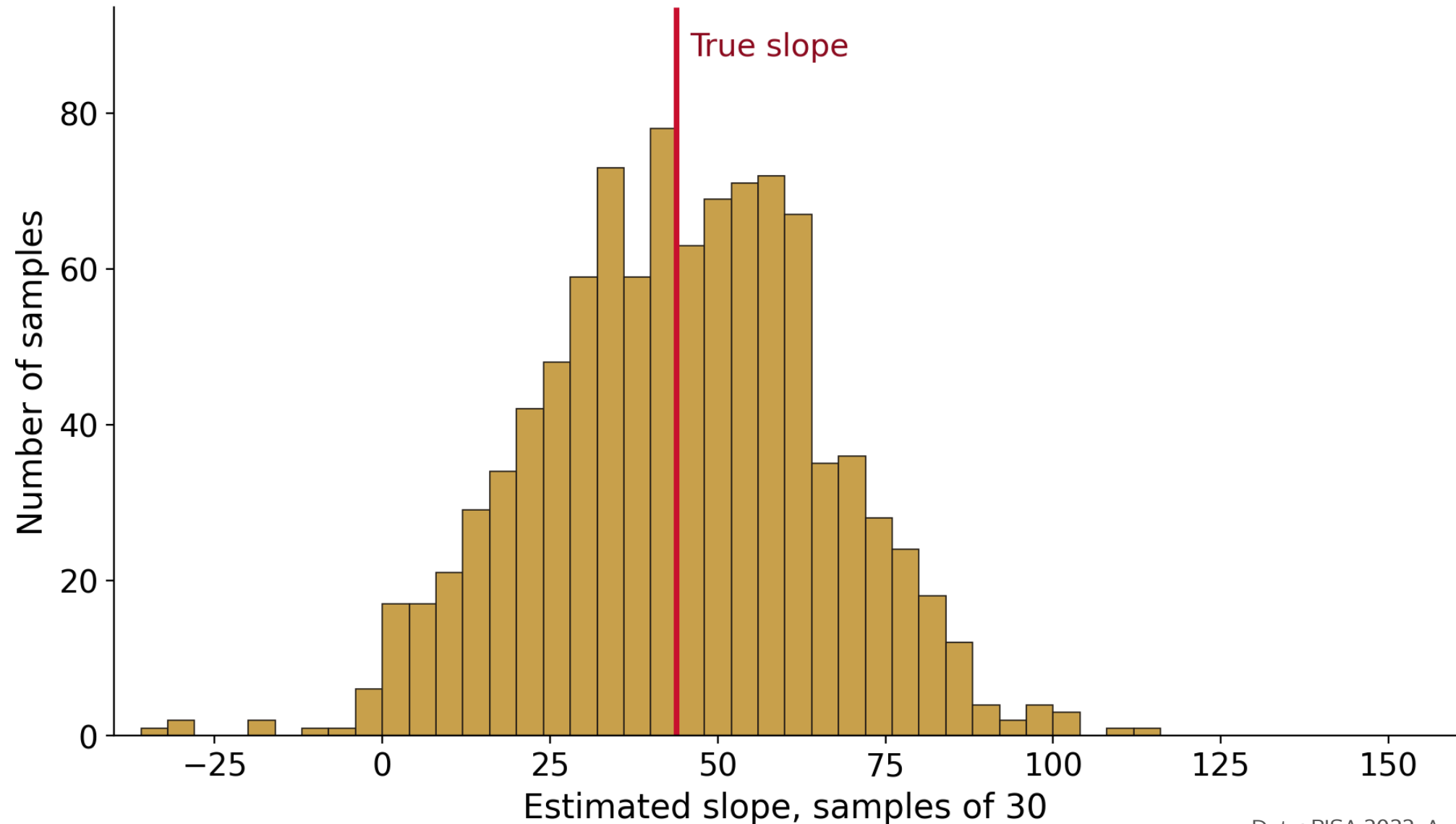
The standard error estimates coeff variability

Standard deviation (across 1,000 slopes): 6.4

Standard error (average, across samples): 6.2

In real work, we have one sample and its standard error

Small samples have more spread-out slopes



Testing null hypotheses

t is the estimate divided by its standard error

$$H_0 : \beta_1 = 0$$

$$t = \frac{\hat{\beta}_1 - 0}{SE(\hat{\beta}_1)} = \frac{78.30}{7.205} = 10.87$$

```
model.params["egm_per_1000"] / model.bse["egm_per_1000"]
```

```
10.866254878788773
```

We can reject the null of a zero pokie slope

```
model.pvalues
```

```
Intercept      7.183543e-02  
egm_per_1000   3.310386e-17  
dtype: float64
```

If $\beta_1 = 0$, how likely is a t at least this far from 0?

Not the probability that $\beta_1 = 0$

The 95% confidence interval for the slope

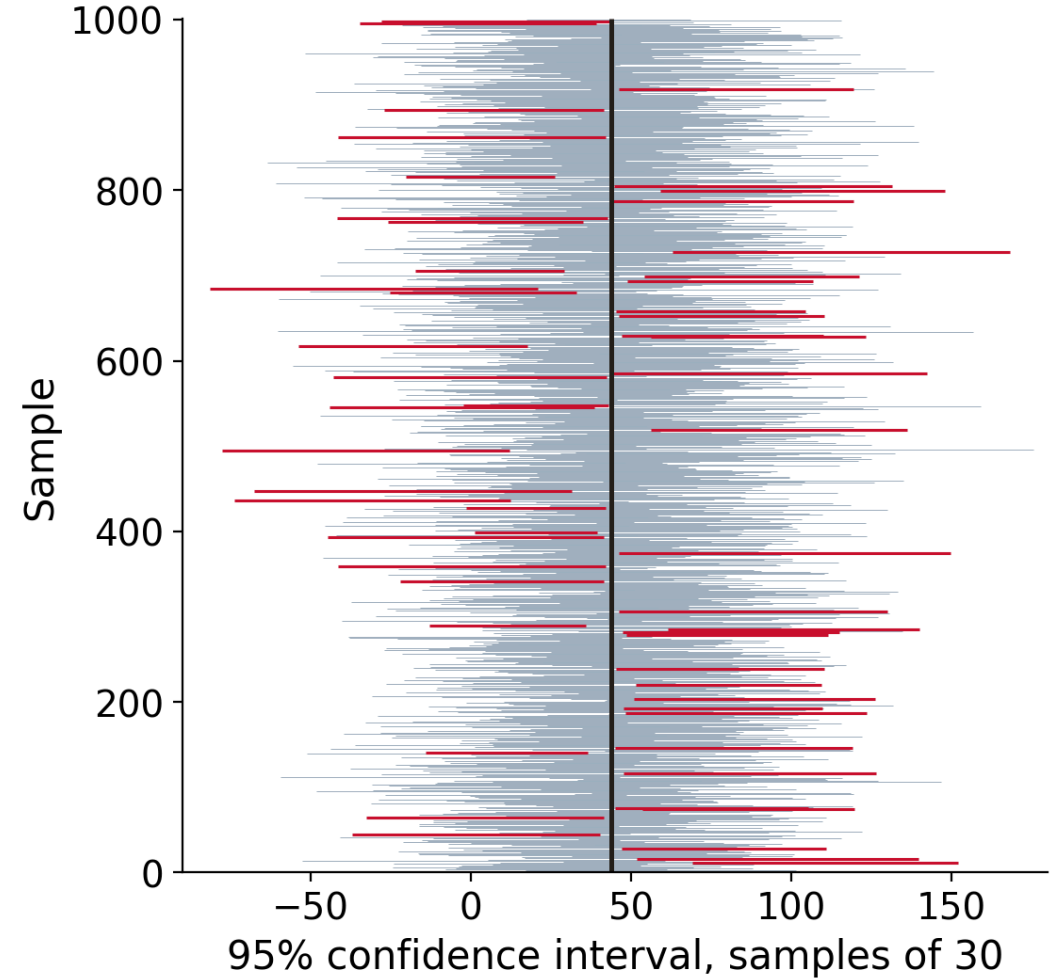
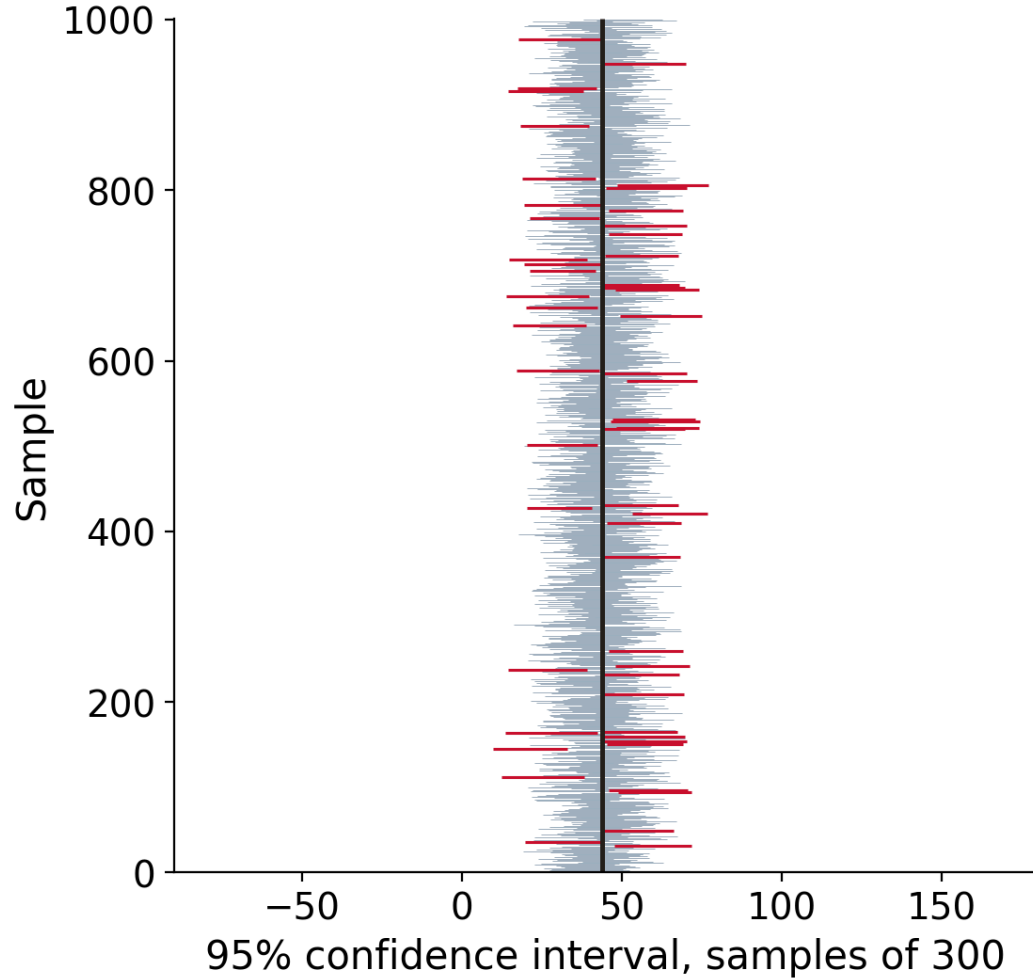
$$\hat{\beta}_1 \pm t^* \times SE(\hat{\beta}_1) = 78.30 \pm 1.99 \times 7.205$$

```
model.conf_int()
```

	0	1
Intercept	-6.471434	148.815821
egm_per_1000	63.948811	92.644775

Across repeated samples, 95% of confidence intervals contain the true slope

Most confidence intervals contain the slope



Data: PISA 2022, Australia (OECD); 1,000 samples of 300 (left) and of 30 (right)

Stars in the table mark p-value thresholds

```
summary_col([model, multi, multi_ue, levels, inter], stars=True, float_format="%.2f",  
            model_names=["Simple", "Dummy", "Percent", "3 levels", "Interaction"])
```

	Simple	Dummy	Percent	3 levels	Interaction
Intercept	71.17*	16.14	-115.93**	13.27	31.94
	(38.99)	(38.74)	(49.52)	(41.58)	(47.19)
egm_per_1000	78.30***	75.87***	72.83***	76.93***	72.33***
	(7.21)	(6.67)	(6.33)	(6.60)	(8.98)
high_ue		134.71***			96.73
		(35.30)			(73.30)
ue_pct			61.18***		
			(11.85)		
C(ue_cat)[T.medium]				28.53	
				(42.57)	
C(ue_cat)[T.high]				167.05***	
				(42.55)	
egm_per_1000:high_ue					7.99
					(13.50)
R-squared	0.61	0.67	0.71	0.68	0.67
R-squared Adj.	0.60	0.66	0.70	0.67	0.66

* p < 0.1 ** p < 0.05 *** p < 0.01 (standard errors)

Do the slopes differ?

Can't reject equal slopes by unemployment

```
inter = ols("exp_per_adult ~ egm_per_1000 * high_ue", data=pokies).fit()
```

	coef	std err	t	P> t	[0.025	0.975]
Intercept	31.9430	47.186	0.677	0.501	-62.056	125.942
egm_per_1000	72.3339	8.976	8.058	0.000	54.453	90.215
high_ue	96.7287	73.302	1.320	0.191	-49.296	242.753
egm_per_1000:high_ue	7.9900	13.496	0.592	0.556	-18.896	34.876

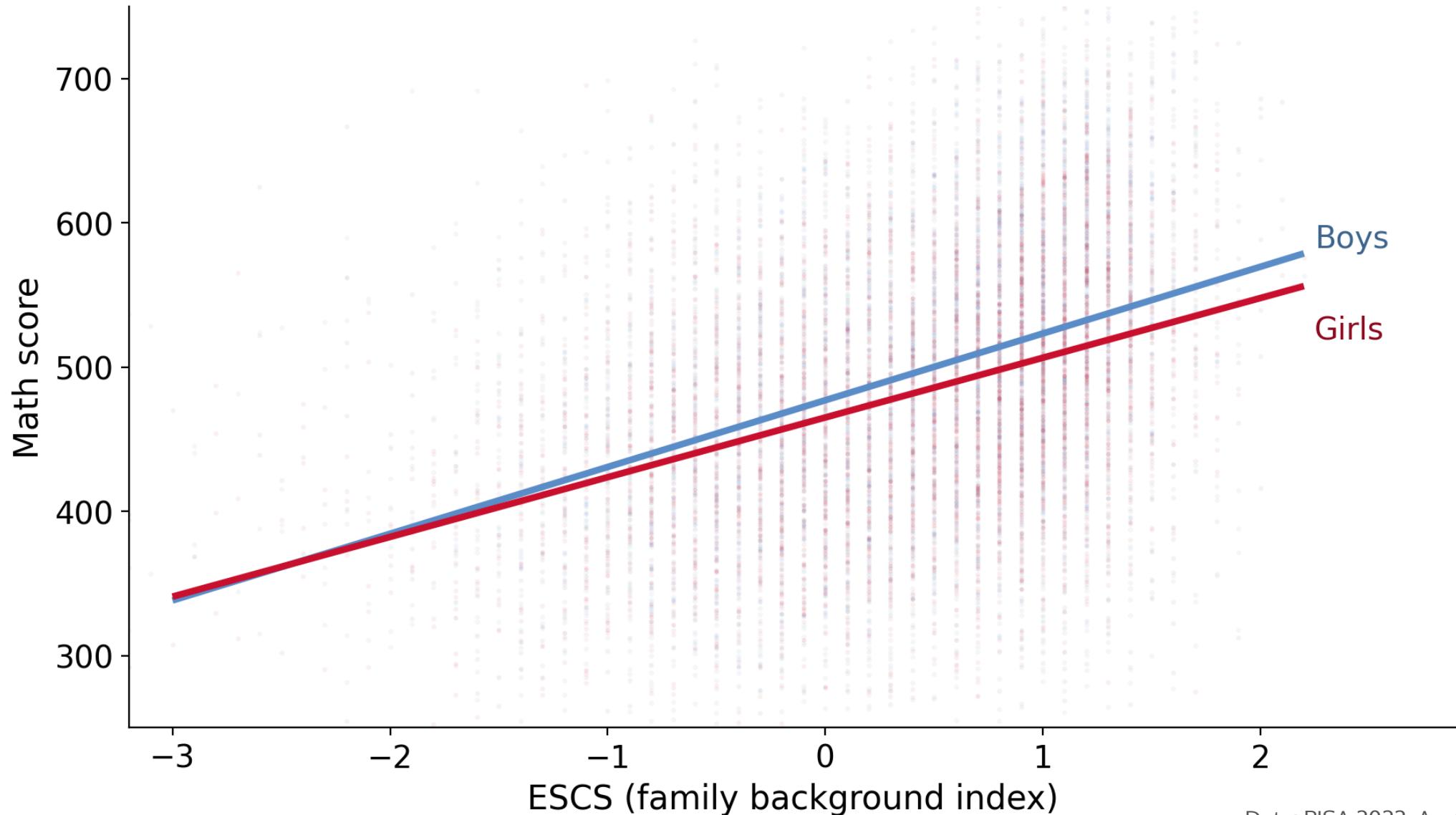
Not rejecting isn't proof of equal slopes

95% confidence interval for the difference in slopes: -18.9 to 34.9

A difference of 0 is inside it

So is a high-unemployment slope \$30 steeper

The ESCS slope is smaller for girls than for boys



Data: PISA 2022, Australia (OECD)

With all the students, we reject equal slopes

```
pisa["female"] = (pisa["gender"] == "female").astype(int)
ols("math ~ escs * female", data=pisa).fit()
```

	coef	std err	t	P> t	[0.025	0.975]
Intercept	477.0806	1.234	386.632	0.000	474.662	479.499
escs	46.2714	1.353	34.209	0.000	43.620	48.923
female	-11.9517	1.767	-6.763	0.000	-15.416	-8.488
escs:female	-4.8629	1.937	-2.510	0.012	-8.661	-1.065

At $n = 300$, we usually can't reject equal slopes

All 12,136 students: p-value 0.012, reject equal slopes

1,000 samples of 300 students: p-value below 0.1 in 12% of them

Small p-values don't make slopes causal

Pokie slope: p-value 0.000

Pokies aren't randomly assigned to LGAs

Omitted variable bias is a separate problem from sampling variation

Next week: joint tests

Essential Concepts: joint tests and the F test

Problem set due Friday 16 October